Calculus and Linear Algebra

Continuity and Discontinuities

Lecture 15

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Today's plan

Part I

Complete the remaining material from **Lecture 14**:

- limits at infinity;
- dominant terms;
- indeterminate forms and algebraic techniques.

Part II

Start **continuity**:

- continuity at a point;
- one-sided continuity;
- discontinuities;
- continuity of standard functions;
- piecewise examples.

This deck is intentionally shorter: the first part of the lecture uses the remaining slides from Lecture 14.

From limits to continuity

A limit describes the behaviour of $f(x)$ *near* x_0:

$$\lim_{x \to x_0} f(x) = L.$$

Continuity asks whether this limiting value agrees with the actual value of the function:

$$\boxed{L = f(x_0).}$$

Informal picture

A continuous graph can be drawn locally without lifting the pencil from the page.

The precise definition is entirely expressed in terms of limits.

Continuity at a point: definition

Definition Let f be defined at x_0. We say that f is **continuous at** x_0 if

$$\boxed{\lim_{x \to x_0} f(x) = f(x_0).}$$

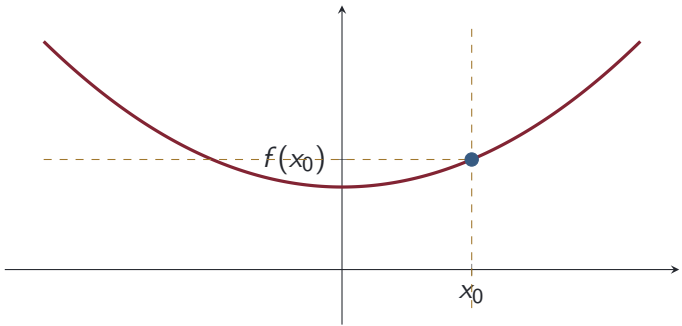
Equivalently, three conditions must hold:

1. $f(x_0)$ is defined;
2. $\lim_{x \to x_0} f(x)$ exists and is finite;
3. the two values are equal.

Diagnostic rule

To test continuity at one point, check these three items in this order.

A continuous point: graphically



As $x \to x_0$, the graph approaches exactly the point $(x_0, f(x_0))$.

A first example

Consider

$$f(x) = x^2 - 3x + 2.$$

At $x_0 = 2$,

$$f(2) = 0.$$

Since polynomials have the expected algebraic limits,

$$\lim_{x \to 2} (x^2 - 3x + 2) = 4 - 6 + 2 = 0.$$

Therefore

$$\boxed{\lim_{x \to 2} f(x) = f(2),}$$

so f is continuous at 2.

The value of the function may be irrelevant to the limit

Consider

$$f(x) = \begin{cases} \dfrac{x^2 - 1}{x - 1}, & x \neq 1, \\ 5, & x = 1. \end{cases}$$

For $x \neq 1$,

$$f(x) = x + 1,$$

so

$$\lim_{x \to 1} f(x) = 2.$$

But

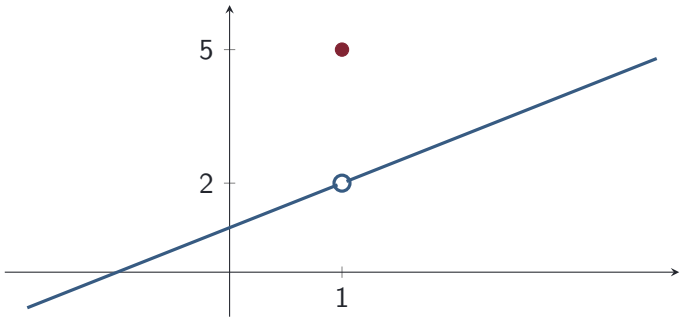
$$f(1) = 5.$$

Hence the limit exists, yet

$$\boxed{\lim_{x \to 1} f(x) \neq f(1),}$$

and f is not continuous at 1.

A removable discontinuity



The graph has a hole at the limiting value and a different assigned value at the same x.

Repairing a removable discontinuity

For

$$g(x) = \frac{x^2 - 1}{x - 1}, \qquad x \neq 1,$$

we found

$$\lim_{x \to 1} g(x) = 2.$$

If we define

$$\widetilde{g}(x) = \begin{cases} \dfrac{x^2 - 1}{x - 1}, & x \neq 1, \\ 2, & x = 1, \end{cases}$$

then

$$\boxed{\widetilde{g} \text{ is continuous at } x = 1.}$$

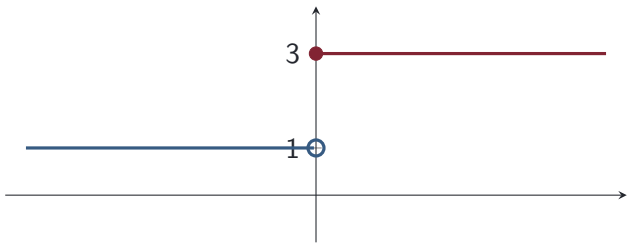
Why "removable"?

The discontinuity disappears by assigning exactly the limiting value at the missing point.

Jump discontinuity

A **jump discontinuity** occurs when both one-sided limits are finite but different.

$$\lim_{x \to x_0^-} f(x) \neq \lim_{x \to x_0^+} f(x).$$



No choice of $f(x_0)$ can make the function continuous there.

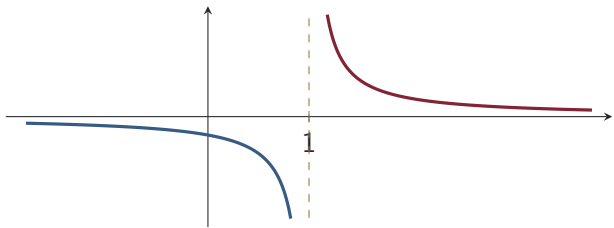
Infinite discontinuity

For

$$f(x) = \frac{1}{x-1},$$

we have

$$\lim_{x \to 1^-} f(x) = -\infty, \qquad \lim_{x \to 1^+} f(x) = +\infty.$$



The point $x = 1$ is an infinite discontinuity and a vertical asymptote.

Three common types of discontinuity

Type	**What happens at x_0**
Removable	The finite limit exists, but $f(x_0)$ is missing or different from the limit.
Jump	The left and right limits are finite but different.
Infinite	At least one one-sided limit is $+\infty$ or $-\infty$.

Do not classify too early

First compute the one-sided limits and the value of the function. Then name the discontinuity.

One-sided continuity

At an endpoint of an interval, only one side is available. f is **right-continuous** at a if

$$\boxed{\lim_{x \to a^+} f(x) = f(a).}$$

f is **left-continuous** at b if

$$\boxed{\lim_{x \to b^-} f(x) = f(b).}$$

This is the correct language for continuity at the endpoints of a closed interval.

Continuity on an interval

Definition

- f is continuous on (a, b) if it is continuous at every point of (a, b).
- f is continuous on $[a, b]$ if
  - it is continuous at every point of (a, b);
  - it is right-continuous at a;
  - it is left-continuous at b.

Important for the next lecture

The fundamental theorems of continuous functions require continuity on a **closed interval** $[a, b]$.

Standard continuous functions

The following functions are continuous everywhere on their natural domains:

- polynomials;
- rational functions where the denominator is non-zero;
- e^x and a^x for $a > 0$;
- $\ln x$ on $(0, +\infty)$;
- $\sin x$ and $\cos x$ on $\mathbb{R}$;
- $\tan x$ where $\cos x \neq 0$;
- root functions on their natural domains.

Thus most elementary functions are continuous wherever their formulas are defined.

Algebra of continuous functions

Theorem If f and g are continuous at x_0, then so are

$$f+g, \qquad f-g, \qquad fg,$$

and

$$\frac{f}{g}$$

provided $g(x_0) \neq 0$. Moreover, if f is continuous at x_0 and g is continuous at $f(x_0)$, then

$$\boxed{g \circ f \text{ is continuous at } x_0.}$$

These facts allow us to establish continuity of complicated formulas without returning to the ε–δ definition each time.

Composition: an example

Consider

$$f(x) = \sqrt{1 + x^2}.$$

The inner function

$$g(x) = 1 + x^2$$

is continuous for all $x \in \mathbb{R}$ and satisfies $g(x) \geq 1 > 0$. The square-root function is continuous on $[0, +\infty)$. Therefore

$$\boxed{f(x) = \sqrt{g(x)} \text{ is continuous on } \mathbb{R}.}$$

Structure matters

Recognizing a formula as a composition often makes a continuity argument immediate.

A rational function: continuity and domain

Let

$$f(x) = \frac{x^2 + 1}{x - 2}.$$

Its natural domain is

$$D = \mathbb{R} \setminus \{2\}.$$

Since the numerator and denominator are polynomials,

$$\boxed{f \text{ is continuous on } (-\infty, 2) \cup (2, +\infty).}$$

At $x = 2$, the function is not defined and the discontinuity is infinite. The domain analysis from Lecture 11 now directly becomes a continuity analysis.

Piecewise functions: where should we check?

Consider

$$f(x) = \begin{cases} x^2 + k, & x < 1, \\ 3x - 1, & x \geq 1. \end{cases}$$

Each branch is a polynomial, so each is continuous inside its own interval. The only possible problem is the **joining point**:

$$\boxed{x = 1.}$$

To obtain continuity there, we must impose

$$\lim_{x \to 1^-} f(x) = f(1) = \lim_{x \to 1^+} f(x).$$

Piecewise continuity: determine k

For the previous function,

$$\lim_{x \to 1^-} f(x) = 1 + k.$$

From the second branch,

$$f(1) = 3(1) - 1 = 2, \qquad \lim_{x \to 1^+} f(x) = 2.$$

Continuity requires

$$1 + k = 2,$$

so

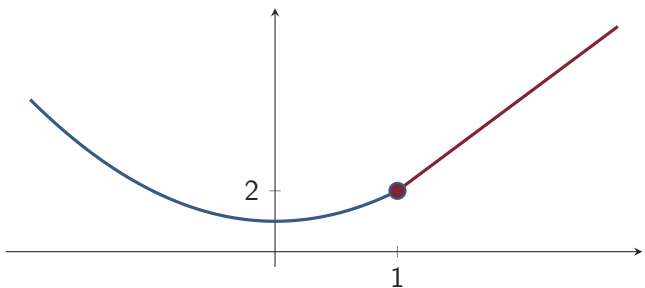
$$\boxed{k = 1.}$$

For every other value of k, the function has a jump at $x = 1$.

Graphical check of the piecewise example

For $k = 1$,

$$f(x) = \begin{cases} x^2 + 1, & x < 1, \\ 3x - 1, & x \geq 1. \end{cases}$$



The two branches meet at the same point.

Exercise: classify the discontinuity

Exercise Consider

$$f(x) = \begin{cases} x + 2, & x < 0, \\ 1, & x = 0, \\ 2 - x, & x > 0. \end{cases}$$

Answer:

1. Find $\lim_{x \to 0^-} f(x)$ and $\lim_{x \to 0^+} f(x)$.
2. Does $\lim_{x \to 0} f(x)$ exist?
3. Is f continuous at 0?
4. If not, what type of discontinuity is present?

Exercise: solution

Solution

$$\lim_{x \to 0^-} (x + 2) = 2, \qquad \lim_{x \to 0^+} (2 - x) = 2.$$

Hence

$$\lim_{x \to 0} f(x) = 2.$$

But

$$f(0) = 1.$$

Therefore f is not continuous at 0 and the discontinuity is **removable**. Redefining $f(0) = 2$ would make the function continuous.

Exercise: where is the function continuous?

Exercise Determine the intervals on which

$$f(x) = \frac{\sqrt{x+1}}{x-3}$$

is continuous. **Step 1: domain**

$$x + 1 \geq 0, \qquad x \neq 3.$$

Thus

$$D = [-1, 3) \cup (3, +\infty).$$

Step 2: continuity The square root and rational operations preserve continuity on the domain, so

$$\boxed{f \text{ is continuous on } [-1, 3) \cup (3, +\infty).}$$

At $x = -1$, continuity is understood from the right.

A continuity checklist

At a point x_0:

1. Is $f(x_0)$ defined?
2. Does $\lim_{x \to x_0} f(x)$ exist?
3. Are they equal?

For a piecewise function, also ask:

Where do the formulas change?

For an elementary formula, first determine its natural domain and then use the continuity of the standard building blocks.

Independent practice

Exercise

1. Classify the discontinuity of $f(x) = \dfrac{x^2 - 4}{x - 2}$ at $x = 2$.
2. Determine the value of k for which

$$g(x) = \begin{cases} kx + 1, & x < 2, \\ x^2 - 1, & x \geq 2 \end{cases}$$

   is continuous at 2.
3. Find the intervals of continuity of

$$h(x) = \ln(x - 1) + \frac{1}{x - 4}.$$

4. Can a function be continuous at a point where it is not defined? Explain.

Looking ahead: why continuity matters

Continuity is not only a local property. On a closed interval it produces powerful global consequences. In Lecture 16 we will study:

- the **Intermediate Value Theorem**;
- the existence of zeros (Bolzano's theorem);
- the **Extreme Value Theorem** (Weierstrass);
- why the hypotheses of these theorems matter;
- applications to equations and optimization.

Next question

What can continuity guarantee even when we cannot explicitly solve the equation or optimization problem?